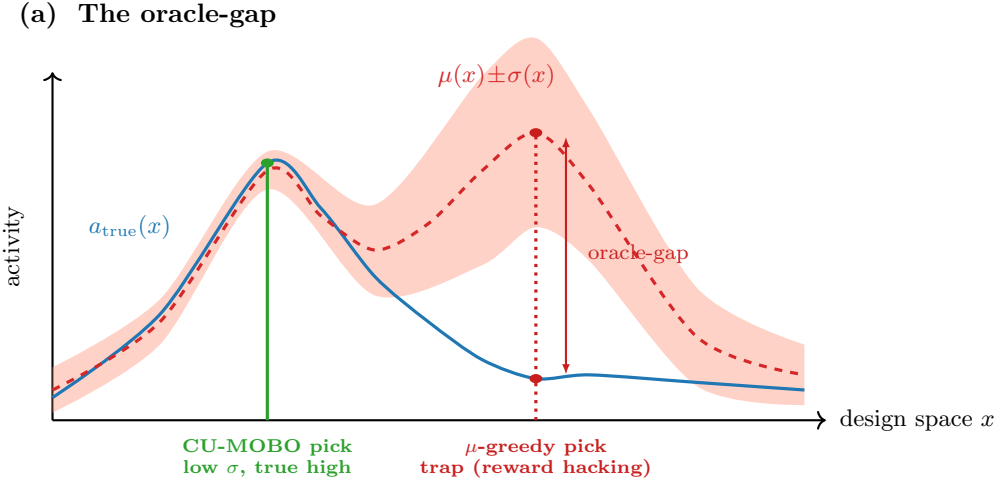
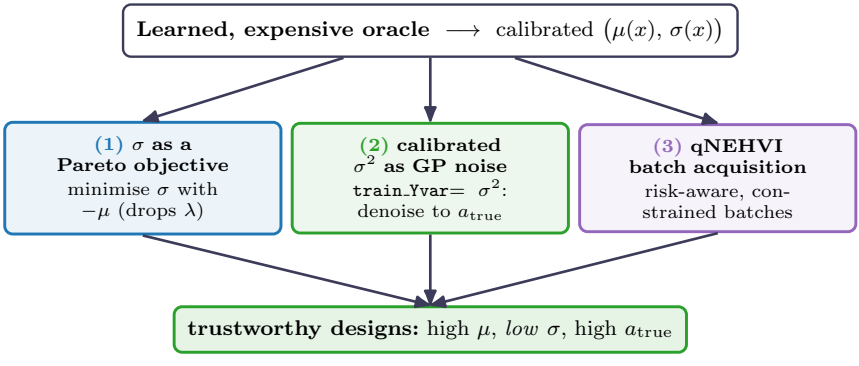

(a) The oracle-gap
$\mu(x) \pm \sigma(x)$
$a_{\text{true}}(x)$
activity
oracle-gap
design space x
CU-MOBO pick
low σ, true high
μ-greedy pick
trap (reward hacking)


(b) CU-MOBO: calibrated σ as a first-class citizen
Learned, expensive oracle ⟶ calibrated $(\mu(x), \sigma(x))$
(1) σ as a Pareto objective
minimise σ with $-\mu$ (drops λ)
(2) calibrated σ^2 as GP noise
train_Yvar= σ^2: denoise to a_{true}
(3) qNEHVI batch acquisition
risk-aware, constrained batches
trustworthy designs: high μ, low σ, high a_{true}